UC San Diego

Predicting Poverty Level in China

Bingru Li, Sunny Xu, Yuchen Wang

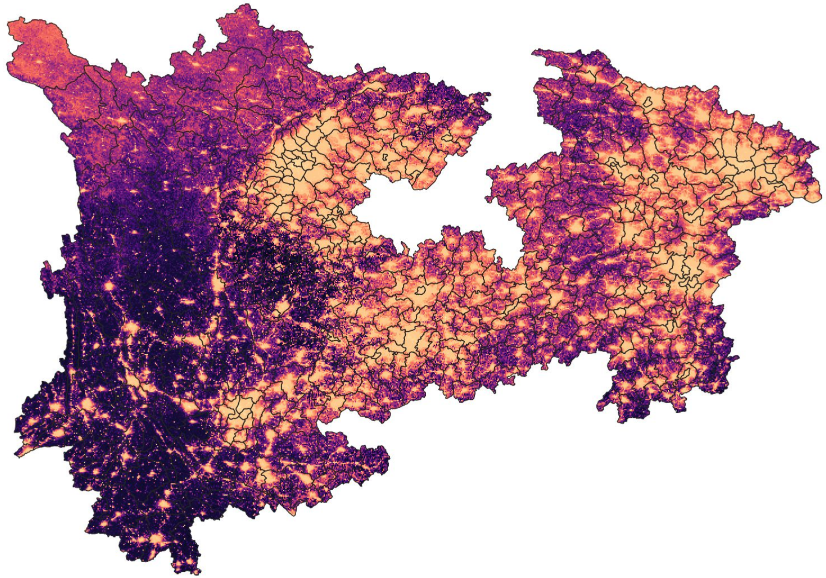
Question

- How to create a more efficient and cost effective way to collect poverty data?
  - Household surveys
    - Expensive, outdated, and biased
    - Some vulnerable groups can be overlooked
  - Solution: Integrating **geospatial data** with existing **household survey data**, we can create an effective method to accurately identifies households in poverty and provides insights into the underlying factors contributing to poverty using **machine learning models**

Question & Background

- Current literature has shown:
  - **Machine learning techniques** have been used in the geospatial domain for a long time.
    - Convolutional Neural Networks (CNN), Classification Tree Analysis (CTA), Random Forest Classifier
  - Satellite data, such as **Night lights** have shown to be a robust and reliable indicator for predicting income/poverty levels with greater accuracy and consistency.

Distribution of Night Light in Study Area (2019)



Data Sources

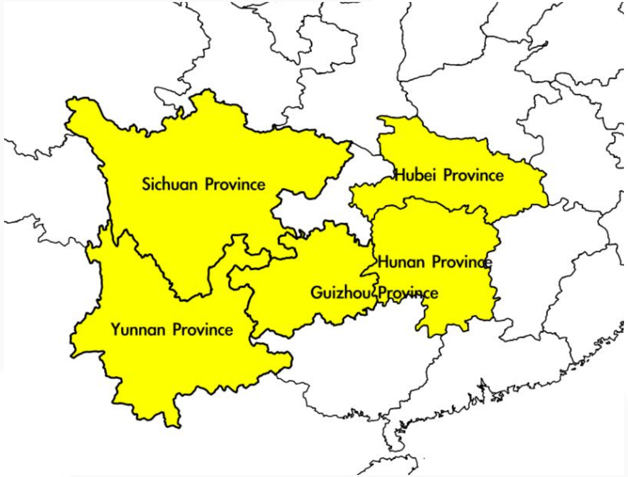
- Household Survey Data
  - China Household Finance Survey (CHFS), 2019
    - 260 counties (districts and county-level cities)
    - Covering about 26,000 households
  - Michael Bauer Research(Regional Market Data) in China, 2022
- Remote-Sensing Data
  - VIIRS nighttime lights data, 2019
  - Normalized Difference Vegetation Index(NDVI) spatial distribution in China, 2019

Packages

- ArcGIS: Obtain county-level demographic data as well as socioeconomic indicators using Geoenrichment
- Pypinyin: Translate Chinese characters into phonetic spellings using pinyin
- Rasterio: Examine the nighttime light data and NDVI data in different counties using the zonal statistics function(zonal_stats)
- Sklearn: Contains machine learning models that can split, train, and test data
- Torch: Contains deep learning models
- Others: Pandas, Geopandas, OS, numpy

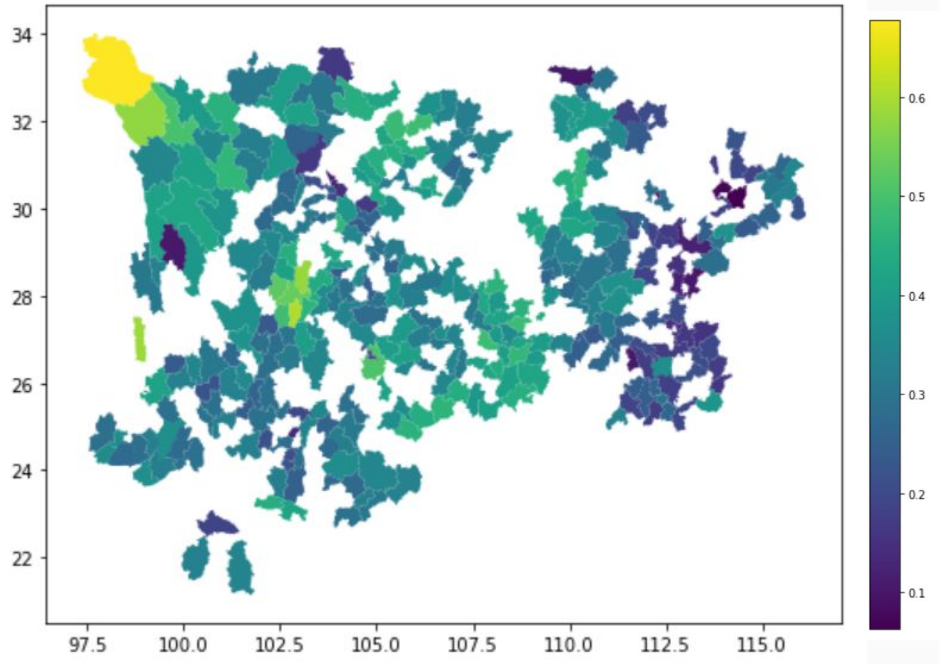
Data Cleaning

- **Number of Observations: 669 -> 501**
- Filter the Household Survey Data to include only the study of interest
  - Study Areas: Yunnan Province, Guizhou Province, Hunan Province, Sichuan Province, and Hubei Province
- Develop a **Poverty Indicator(Response Variable)**:
  - Define the poverty rate as the percentage of individuals living in absolute poverty (annual income < 4000 CNY) within counties.
  - Assign a value of 1 if the county's poverty rate is within the upper-25% percentile of the province poverty rate.
- Ensure consistent display of counties across all datasets.
- Merge all the datasets together



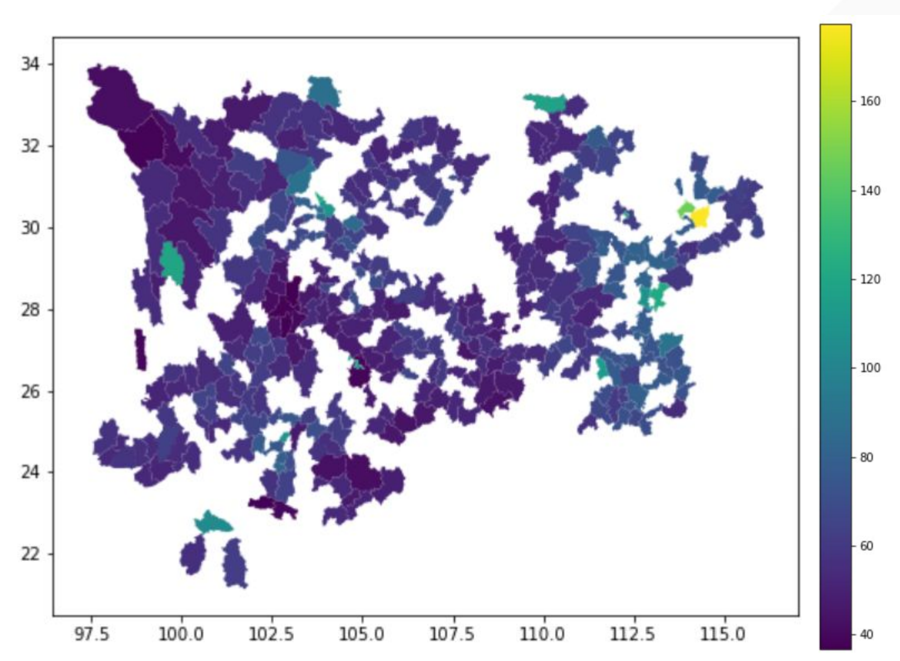
Descriptive Statistics

Distribution of Poverty Household percentage
(calculated from survey data)

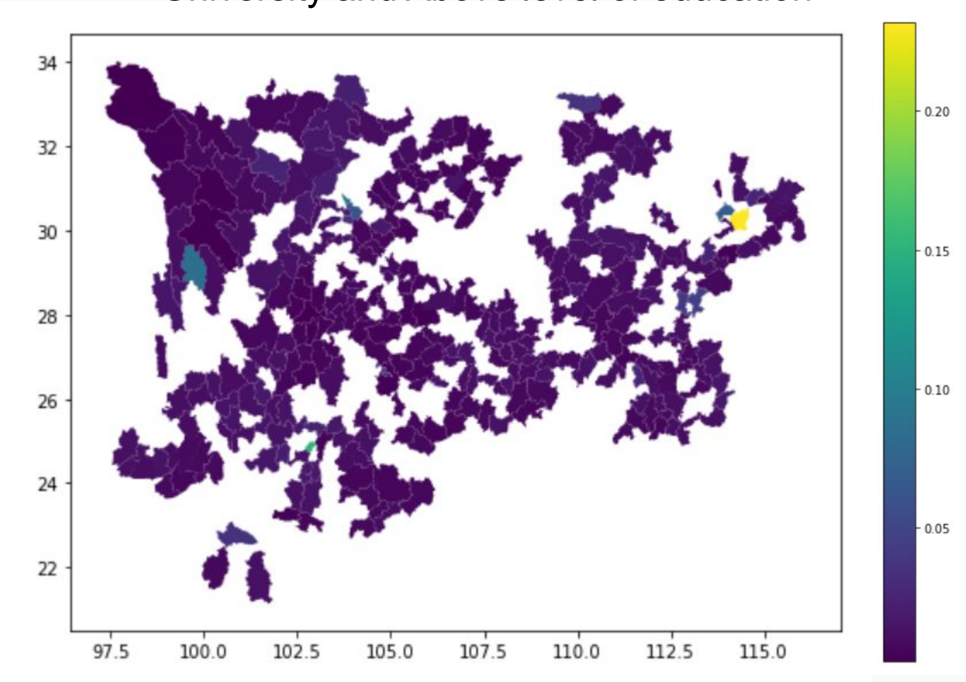


Descriptive Statistics (continued)

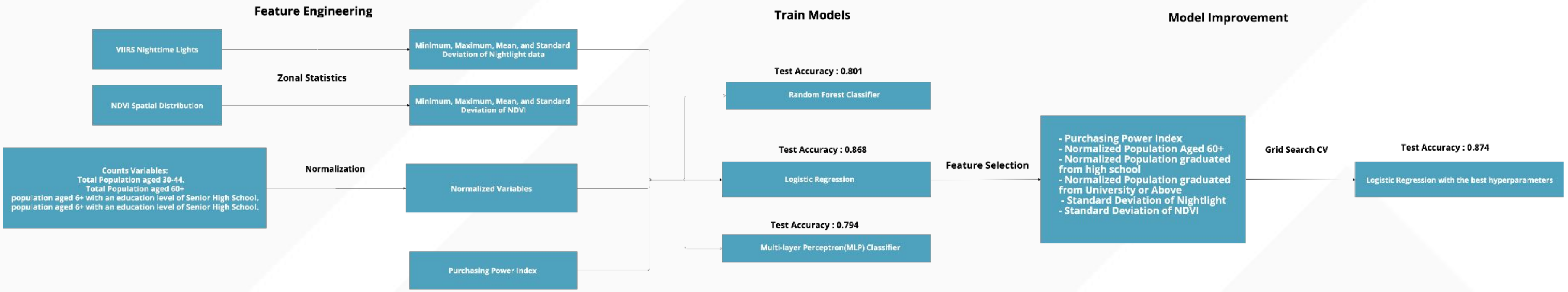
Distribution of Purchasing Power Index



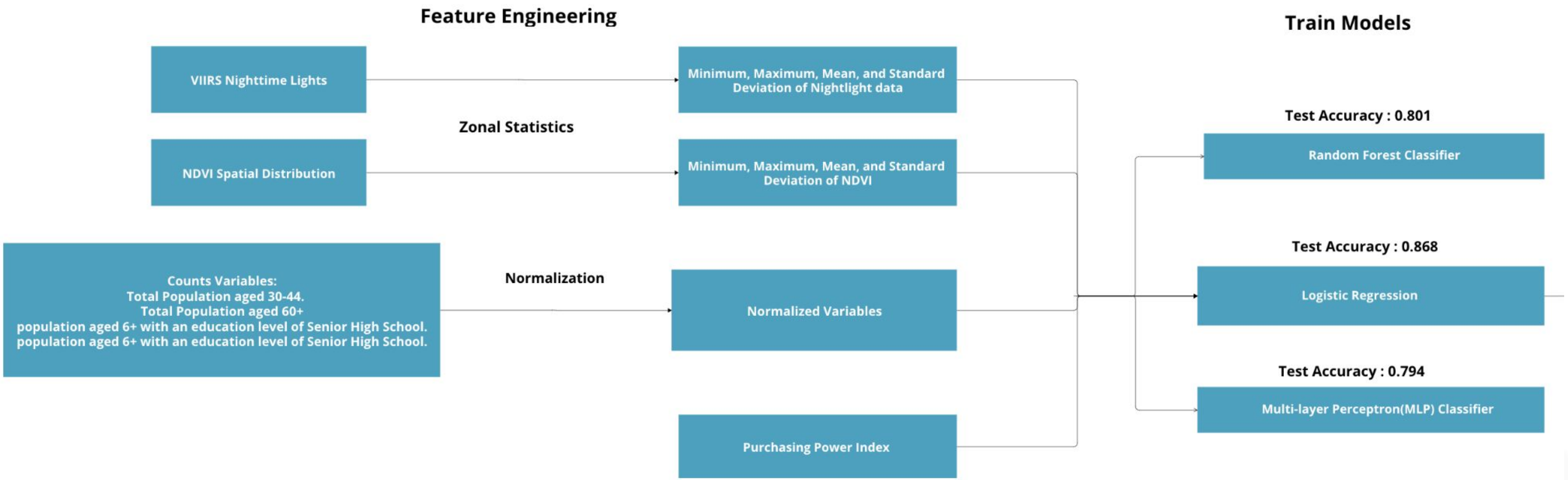
Distribution of population percentage received University and Above level of education



Modeling Poverty

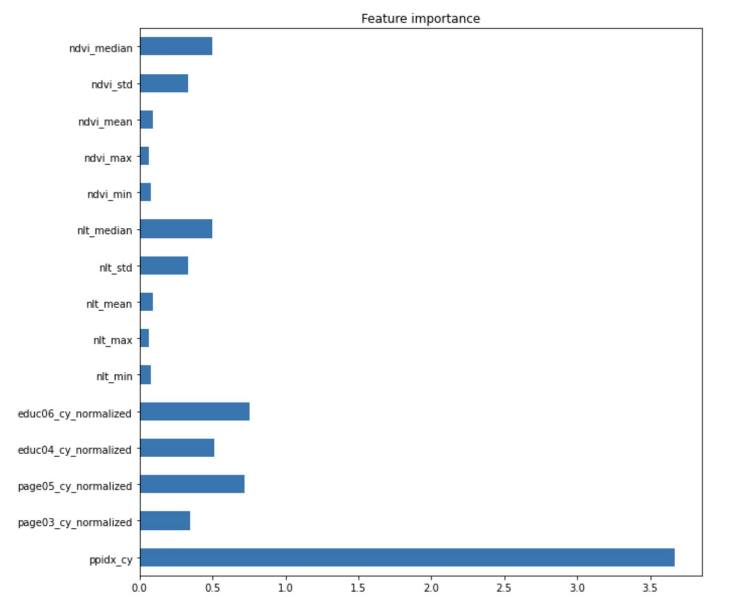


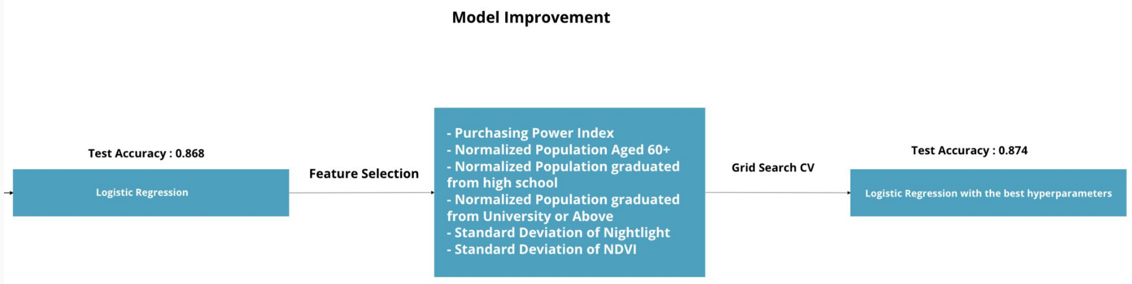
Modeling Poverty



Modeling Poverty

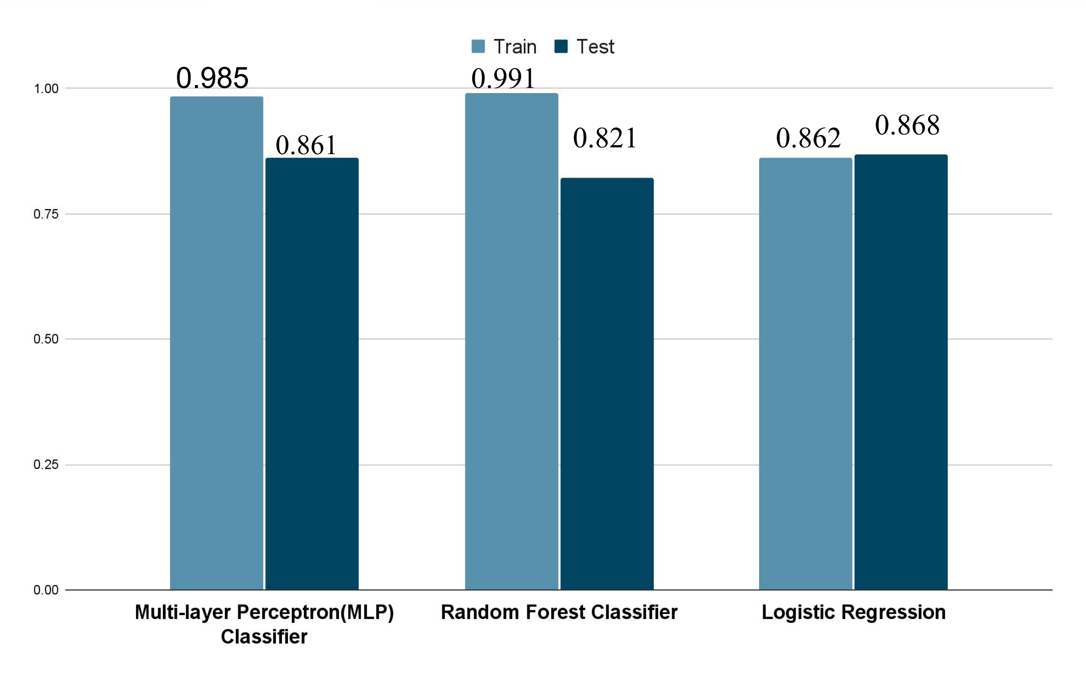
Feature Importance





Results & Conclusions

- Achieved a model accuracy score of 0.868 using logistic regression.
- Purchasing Power Index is most important in predicting poverty



Future Work

- Limitation: Imbalanced data with a majority of 0s and few 1s for the poverty indicator.
  - Solution: Implemented SMOTE (Synthetic Minority Over-sampling Technique) to balance the dataset.
- Potential Steps for Improvement:
  - Incorporate road network analysis to understand patterns and improve predictions.
  - Application to the Country as a Whole
    - Address concerns about provincial variations and adapt the model to different regions
- Expected Utilization: Policymakers, government agencies, and organizations involved in poverty alleviation and resource allocation.

Q & A